

Discrete Shape Space and Integer Logarithmic Lattices

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Abstract

Many physical and geometric observables are inherently multiplicative: their ratios, not their absolute values, carry intrinsic meaning. When expressed in logarithmic coordinates, an overall additive shift is physically redundant. Removing this redundancy defines a quotient space we call *shape space*. We show that for quadratic descriptions of rank n , the number of independent scale-free deformation directions is exactly $n^2 - 1$, the dimension of the traceless endomorphism space $\text{End}^{\text{tr}=0}(V)$. For $n = 3, 4, 5$ this yields 8, 15, 24. We establish this as a formal proposition with proof, note its coincidence with adjoint representation dimensions of $\text{SU}(n)$, and discuss how golden-ratio-saturated geometries provide examples in which quadratic invariants remain within the rank-2 integer module $\mathbb{Z}[\varphi]$.

1 Introduction

Many quantities of physical and geometric interest span many orders of magnitude. Their ratios, not their absolute values, carry the meaningful information. This suggests working in logarithmic coordinates, where multiplicative relationships become additive. The space of logarithmic coordinates carries a redundancy: an overall additive constant corresponds to a simultaneous rescaling of all quantities. Factoring out this redundancy leads to *shape space*.

In this paper we study the structure of shape space for quadratic data. Our central result (Proposition 3.1) is that the space of independent scale-free deformation directions for a rank- n quadratic description has dimension $n^2 - 1$, following from the rank-nullity theorem applied to the trace map on endomorphisms.

2 Logarithmic Coordinates and Shape Space

Definition 2.1 (Shape space). The *shape space* associated with N positive quantities is the quotient $\mathcal{S}_N = \mathbb{R}^N / \mathbb{R}\mathbf{1}$, where $\mathbb{R}\mathbf{1} = \{(t, \dots, t) : t \in \mathbb{R}\}$. It has dimension $N - 1$.

3 The $n^2 - 1$ Count

Infinitesimal deformations of quadratic data on an n -dimensional space are represented by endomorphisms $T: V \rightarrow V$. Uniform rescalings $T = \lambda I$ are shape-redundant and must be quotiented out.

Proposition 3.1 (Dimension of shape-space deformations). *The space of independent scale-free deformation directions of a rank- n quadratic description is*

$$\dim \text{End}^{\text{tr}=0}(V) = n^2 - 1,$$

where $\text{End}^{\text{tr}=0}(V) = \{T \in \text{End}(V) : \text{tr } T = 0\}$.

Proof. The trace map $\text{tr}: \text{End}(V) \rightarrow k$ is surjective with kernel $\text{End}^{\text{tr}=0}(V)$. By the rank-nullity theorem, $\dim \text{End}^{\text{tr}=0}(V) = \dim \text{End}(V) - 1 = n^2 - 1$. The canonical splitting $\text{End}(V) = \text{End}^{\text{tr}=0}(V) \oplus k \cdot I$ confirms that $\text{End}^{\text{tr}=0}(V)$ is exactly the scale-free quotient. $\square$

Corollary 3.2. *For ranks $n = 3, 4, 5$ the independent deformation count is 8, 15, 24.*

Remark 3.3 (Relation to $\text{SU}(n)$ adjoint representations). The integers $n^2 - 1$ are also the dimensions of the adjoint representations of $\text{SU}(n)$: $\dim \mathfrak{su}(3) = 8$, $\dim \mathfrak{su}(4) = 15$, $\dim \mathfrak{su}(5) = 24$. The Lie algebra $\mathfrak{su}(n)$ consists of skew-Hermitian traceless $n \times n$ matrices, the complex analogue of the real traceless endomorphisms here. The present derivation reaches the same count from scale-free shape space without invoking any symmetry assumption [3, 4].

4 Golden-Ratio Closure and Quadratic Invariants

Many discrete geometries involve the golden ratio [1] $\varphi = (1 + \sqrt{5})/2$, which satisfies $\varphi^2 = \varphi + 1$. The set $\mathbb{Z}[\varphi] = \{a + b\varphi : a, b \in \mathbb{Z}\}$ is a commutative ring (the ring of integers of $\mathbb{Q}(\sqrt{5})$ [2]).

Proposition 4.1 (Quadratic closure of $\mathbb{Z}[\varphi]$). *Let $T \in M_n(\mathbb{Z}[\varphi])$. Then $\text{tr}(T^2) \in \mathbb{Z}[\varphi]$. More generally, all polynomial invariants of T with integer coefficients lie in $\mathbb{Z}[\varphi]$.*

Proof. $\mathbb{Z}[\varphi]$ is a commutative ring closed under addition and multiplication. The (i, j) entry of T^2 is $\sum_k T_{ik} T_{kj} \in \mathbb{Z}[\varphi]$, and $\text{tr}(T^2) = \sum_i (T^2)_{ii} \in \mathbb{Z}[\varphi]$. The argument for general polynomial invariants is identical. $\square$

Remark 4.2 (Scope of Proposition 4.1). Proposition 4.1 establishes that if deformation data lies in $\mathbb{Z}[\varphi]$, then quadratic invariants also lie in $\mathbb{Z}[\varphi]$. It does not select the values $n = 3, 4, 5$; those come from Proposition 3.1. The role of $\mathbb{Z}[\varphi]$ is algebraic closure, not geometric selection.

5 Orientation and Signed Directions

Definition 5.1 (Orientation-reversed deformation). Let $T \in \text{End}^{\text{tr}=0}(V)$. The *orientation-reversed* deformation is $-T \in \text{End}^{\text{tr}=0}(V)$. In a Hermitian setting one may use the adjoint $T^\dagger$, which reverses orientation while preserving norm.

Signed integer coefficients in logarithmic coordinates have direct geometric meaning: $+k$ represents k steps forward along a deformation axis, $-k$ represents k steps in reverse.

6 Scope and Generalization

The derivation in Section 3 uses only finite dimensionality of $\text{End}(V)$ and surjectivity of the trace map. It does not depend on any specific group, representation, or geometric realization. For other discrete geometries the role of $\mathbb{Z}[\varphi]$ in Proposition 4.1 is replaced by the appropriate ring of integers of the relevant quadratic field; the formula $n^2 - 1$ remains universal.

7 Conclusion

We have shown that $n^2 - 1$ arises as the dimension of the space of independent scale-free deformation directions for a rank- n quadratic description, proved by rank-nullity alone. For $n = 3, 4, 5$ this gives 8, 15, 24. Golden-ratio-saturated geometries provide examples where all quadratic invariants remain within $\mathbb{Z}[\varphi]$ by ring closure. The construction is independent of any specific physical model.

References

- [1] H. S. M. Coxeter. *Regular Polytopes*. Dover, 1973.
- [2] J. Neukirch. *Algebraic Number Theory*. Springer, 1999.
- [3] J. G. Ratcliffe. *Foundations of Hyperbolic Manifolds*. Springer, 2006.
- [4] H. Weyl. *The Classical Groups*. Princeton University Press, 1946.